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An Evaluation of Different Annotation Approaches for YOLOv8 Instance Segmentation in UAV Imagery: A Case Study on the UAV-BD Dataset

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Abstract. This paper presents a comprehensive analysis of the impact of distinct annotation approaches, namely bounding boxes (B-Box) and masks, on the accuracy of YOLOv8-seg, a state-of-the-art instance segmentation model, when applied to unmanned aerial vehicle (UAV) imagery. Our investigation centres on the UAV-BD dataset, a challenging and diverse dataset designed for UAV applications. Through meticulous comparisons, we assess the performance of YOLOv8-seg using mAP50 and mAP50-95 metrics under different annotation approaches, providing an understanding of the efficacy of each annotation method. Our study yields intriguing findings: while mAP50 scores exhibit comparability between bounding box and mask annotations, a distinctive divergence surfaces in the mAP50-95 metric. The mask annotation approach consistently outperforms bounding boxes, suggesting its dominance in effectively handling instances with higher IoU thresholds under all the backgrounds (i.e., sand, lawn, bush, land, step, ground and playground). The enhanced performance of mask-based annotations, particularly at higher IoU thresholds. This research provides valuable insights that can inform decision-making in digital twin applications and other domains reliant on accurate UAV-based object detection, fostering advancements in technology and application domains.

Keywords: Annotation Approach, Instance Segmentation, Riparian Waste Detection, UAV

1 Introduction

Unmanned aerial vehicle (UAV) imagery plays a pivotal role in diverse applications, ranging from surveillance to environmental monitoring. Accurate object detection and segmentation in UAV imagery are critical for extracting meaningful information and enhancing the capabilities of applications such as digital twin technology. In this context, the choice of annotation approaches significantly influences the performance of in-instance segmentation models. This paper delves into a detailed examination of the impact of different annotation methods, specifically bounding boxes (B-Box) and masks, on the accuracy of YOLOv8-seg, a cutting-edge instance segmentation model, when applied to UAV imagery. The UAV-BD dataset serves as the focal point of our investigation, providing a challenging and diverse set of scenarios that mimic real-world UAV applications. Designed to encompass various environments, including sand, lawn, bush, land, step, ground, and playground, the UAV-BD dataset reflects the complexity of UAV imagery encountered in practical scenarios. Our research adopts a meticulous comparative approach, employing mean Average Precision at IoU 50 (mAP50) and mean Average Precision between IoU 50 and 95 (mAP50-95) metrics to assess the performance of YOLOv8-seg under distinct annotation approaches. This evaluation aims to elucidate the efficacy of each annotation method and uncover their impact on the model's accuracy.

2 Work flow

Training UAV-BD dataset (train/valid) with mask annotations, inferring the UAV-BD dataset (test part) and getting the results derived from both the mask- and bounding-box- based standards (i.e., True Label and evaluation approach). To understand the accuracy derived from the annotation difference, the authors compared the Precision, Recall, mAP50, mAP50-95 and F1 score (IoU=0.45) between mentioned different two annotation approach (i.e., mask-based value minus bounding-box-based value) under various backgrounds.

3 Conclusion

As shown in the Figure 1, the findings of our study reveal intriguing nuances in the performance of YOLOv8-seg under different annotation approaches. Notably, while mAP50 scores exhibit comparability between bounding box and mask annotations, a distinct divergence emerges in the mAP50-95 metric. The mask annotation approach consistently outperforms bounding boxes, showcasing its dominance in effectively handling instances with higher IoU thresholds across various backgrounds. Considering the reasons of the evaluation-index-based difference, Figure 2 has depicted that the backgrounds in the bounding box have been trained in the model, and the bottle-based overlapping area probably have been reduced.

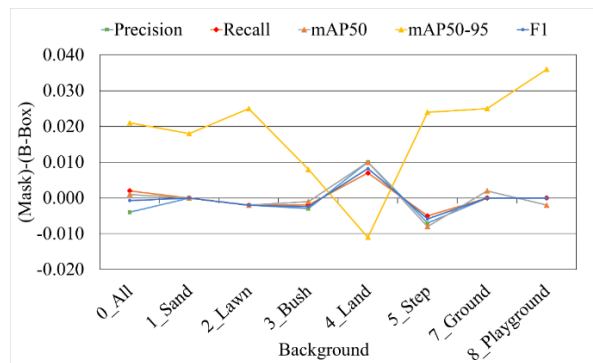


Figure 1 Comparison of the F1 score (IoU=0.45), mAP50 and mAP50-95 between the mask- and bounding-box- based standards.

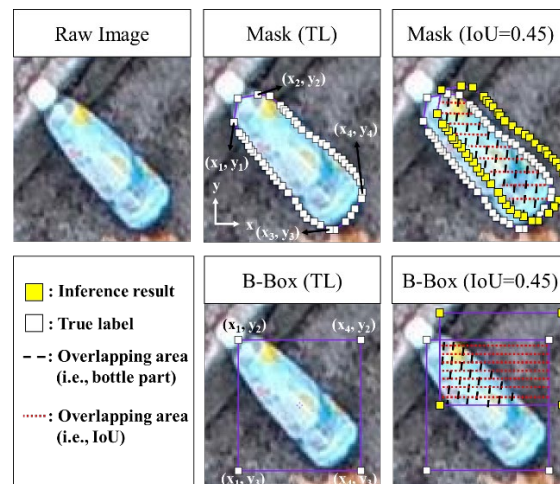


Figure 2 True Label (TL) and evaluation approach difference.

Reference

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